

HEMANT SHARMA

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Experience

Clientell | Senior Software Engineer

October 2025 – Present

Python, FastAPI, PostgreSQL, Redis, AWS, ECS, Terraform, GCP, OpenTelemetry

Bengaluru, Karnataka

- Own the **agent execution path** from tool call to stream delivery; drove tool-call latency from **~4s to ~800ms**.
- Designed the Salesforce org indexing platform and took it through founder review, then built it as an **orchestrator-worker system on Pub/Sub**: ~28k work items per run, workers autoscaling to 100 Cloud Run instances, fan-out bounded by customer API quota.
- Led the migration of all **9 services from GCP to AWS** (Cloud Run to ECS on Fargate, Pub/Sub to SQS/SNS, Cloud SQL to RDS, Memorystore to ElastiCache); chose Terraform for the rebuild and owned the cutover plan and execution. Completed in **7 days with 1 hour of downtime and no rollback**.
- Took over a p99 regression that had **escalated past initial triage** (~1.4s against a ~400ms baseline at flat RPS, no hot path in the profile) and root-caused it to **SQLAlchemy sessions held across an async boundary**, so pool checkouts outlived request scope. Fixed the session lifecycle, sized the pool to measured service time, added a saturation alert; **p99 has held at baseline since**.
- Made SSE streams survive bursty upstream emission (5 to 15 second OpenAI token bursts) with a bounded queue that drains in fixed batches, throttles delivery, and retries transient faults. Forced-kill failures went from **~4/day to none across the last ~500 streams**; wrote the runbook and handed ownership to two engineers.
- Built **observability from zero across all 9 services** and set the SLOs the team now runs against. Median time-to-detection went from **over a day to under 10 minutes**; no incident has reached us via a customer since.
- **Mentor two engineers** on backend design; run the weekly design review that gates changes to the execution path.

Fae | Backend Engineer

January 2025 – October 2025

FastAPI, Go, PostgreSQL, Docker, AWS ECS, GitHub Actions

Bengaluru, Karnataka

- Rewrote a legacy NestJS service into a two-service FastAPI codebase with dependency injection across API, service, and data layers, cutting the 8-10 story points of refactoring the old structure forced each sprint.
- Closed permission gaps on the internal ground-ops platform with **ABAC authorization** and an audit log that ties every operation to the acting identity; monthly access and performance audits now run against it.
- Moved slow external payment calls off the critical path and tightened transaction boundaries; payment execution latency halved from **~5s to ~2.5s** and timeout failures stopped.

Gida Technologies | Software Engineer

October 2022 – November 2023

Python, Node.js, Google BigQuery, PostgreSQL

Bengaluru, Karnataka

- Built a daily ETL pipeline that loads **~600k health insurance records** into BigQuery with source-level segregation and PGP encryption, meeting Ayushman Bharat Digital Mission (ABDM) regulatory requirements.
- Automated recurring analyst reporting (Python jobs orchestrated from Node.js) and cut **~20 hours/week** of manual work.

Projects

Redis-Compatible In-Memory Server | Go

- Built a **Redis-protocol-compatible** server supporting GET/SET, HGET/HSET, and ZADD, with automatic key expiration and durability to SQLite.

Consistent Hashing Library | Go

- Implemented a consistent hash ring with 20 virtual replicas per node and $O(\log N)$ lookups via binary search over sorted hashes. Hashing is pluggable (CRC32/FNV-1a), reads and writes are thread-safe, and the ring replaces atomically in full.

Distributed MapReduce (MIT 6.5840) | Go, RPC

- Built a coordinator-worker MapReduce system with exactly-once task semantics and crash recovery; heartbeat monitoring and idempotent operations handle worker failures under concurrent execution.

Technical Skills

Languages: Python, Go, TypeScript, SQL

Backend & Systems: FastAPI, REST APIs, gRPC, WebSockets, Distributed Systems, Concurrency

Infrastructure: PostgreSQL, Redis, BigQuery, OpenTelemetry, Docker, Terraform, GitHub Actions, AWS, GCP, Linux

Education

Manipal Institute of Technology

July 2017 – June 2021

Bachelor of Technology

Manipal, Karnataka